PHAM ANH DUY

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Summary

Anh-Duy Pham is a machine learning researcher driven by a deep fascination with generative deep learning and computer vision. He is particularly passionate about developing AI methods that enable intelligent image and signal processing, optical sensing systems, and edge-deployed solutions for real-world monitoring tasks. His genuine interests lie in creating explainable, efficient models that bridge simulation, data modelling, and practical perception challenges, with a strong curiosity for applying these technologies to advance scientific understanding and sustainable innovation.

Professional Experience

Doctoral Researcher / PhD Student, *Würzburg Univsersity* 04/2025 – 06/2026
Generative AI for Operations Research

Doctoral Researcher / PhD Student, *Osnabrück Univsersity* 12/2023 – 03/2025
Informed Machine Learning for Agriculture

Student Research Assistant, *Fraunhofer HHI* 09/2022 – 09/2023
Berlin
- Machine learning and Computer Vision for palm vein recognition
- Introduce inductive bias from StyleGAN components into autoencoders for better Deepfakes (Master thesis)

Student Research Assistant, *Chair of Macroeconomics - Ruhr-Universität Bochum* 10/2021 – 08/2022
Analyze effects of climate change on Economics with Agent-based Modelling

Student Research Assistant, *Fraunhofer SCAI* 07/2021 – 08/2022
Machine learning on CAE simulation Sankt Augustin

Werkstudent, Software Engineer, *Autodesk* 02/2020 – 06/2021
Bonn
- Spearhead front-end web development for Autodesk Viewer – a spare web application to view and share any Autodesk files
- Elaborate unit tests and integration tests accordingly
- Actively search and fix bugs

Volunteering Machine Learning Researcher, *Intel Vietnam* 11/2020 – 2022
Vietnam
- Automate QA processes with image processing and object detection techniques such as Scaled-Yolo-v4 in an end-to-end application
- Reach the speed of 4 images per second and counting (for offline processing) with accuracy of 97%

Freelance Data-Forecasting Researcher, *Vietnam Electricity* 02/2019 – 2021
Vietnam

- Navigate thorough analyses of electricity consumption data in Ho Chi Minh city to provide future demands to the planning department
- Advance development of a new statistical time series data filtering method to achieve much exact short-term electric load data (decrease maximum of 1% Mean Absolute Percentage Error compared to using raw data and the results are summarized in the listed papers)

Research and Teaching Assistant for Electrical Engineering and Information Technology, *Vietnamese-German University* 10/2017 – 07/2019
Vietnam

- Administered assignments and tutorials for the module Digital Signal Processing
- Conducted critical research in the field of machine learning, artificial intelligence, and medical image processing
- Utilized machine learning in functional Magnetic Resonance Imaging (fMRI) for group-level dynamic brain connectivity analysis
- Pioneered the VGU-funded research project "Aerial-view traffic image segmentation using deep learning" which consisted of implementation of state-of-the-art traffic image segmentation module and its integration to a self-built Unmanned Arial Vehicle
- Directed students thorough projects including their bachelor theses and major projects

Education

Master of Science (M.Sc.) in Autonomous Systems, *Hochschule Bonn-Rhein-Sieg* 2023

Master of Science (M.Sc.) in Global Production Engineering and Management, 2017
Technical University of Berlin

- Garnered a wealth of practical experience in manufacturing and factory planning, production technology, additive manufacturing, engineering design, systematic product development methods-time measurement, automation technology, quality management, analytical reasoning, critical thinking, negotiation, manufacturing process planning, lean thinking, eliminating production wastes, quality control (FMEA), product designing and planning, communication skills and teamwork
- Master thesis: builds a roadmap for developing a virtual learnstrument regarding to Industry 4.0. By researching the demands in the future industry and qualifications requirements from companies to the current workforce, teaching contents are integrated into a virtual learning environment built with Unity.

Bachelor of Engineering in Electrical Engineering and Information Technology, 2015
Frankfurt University of Applied Sciences, Germany

- Cultivated hands-on experience in electric circuits, electronics and power engineering, software Engineering, telecommunication techniques, robotics, machine learning, analytical thinking, electronics circuit designing, robotics programming machine learning, image processing and pattern recognition, deep learning, data analysis, control engineering, and software development
- Bachelor thesis: focused on applying conventional machine learning technique Support Vector Machines for motorbike license plate recognition and the algorithm was successfully implemented and validated on the personally designed drone

Core Competencies

Mathematical ML & AI
Sheaf theory, algebraic topology (persistent homology), graph-based modeling, and multimodal large models

Programming
Python, Javascript, C++

Agentic AI
Ontology design, prompt engineering with Langgraph, and GraphRAG

Frameworks
PyTorch (advanced weight manipulation), TensorFlow, Jax, scikit-learn

Data Science
Feature engineering, data wrangling, cleaning, visualization, and annotation

Software Engineering
CI/CD (Jenkins, GitHub Actions, Travis CI), unit & integration testing (pytest, Jest, Cypress)

Certifications

- The contest of SEA Makerthon – Design a world with zero wastes (2nd place national)
- The contest Tikkun Olam Makers – Helping disabled children (Volunteer – candidate)
- Deep Learning Specialization on Coursera

Project Experience

Explainable AI for Multivariate Time Series Classification 08/2021

- Project goal: Pull faithful explanations from a trained classification model with multivariate time series
- Project outcome: Based on my own ranking scheme. CNN-based explanations are better than attentions in terms of extracting faithful explanation.
- Project methods: Using feature maps visualization methods such as Class Activation Mapping for Convolutional Neural Network-based classification models and attention neural models for Recurrent Neural Network-based models; being inspired by two-phase or two-level deep learning models that support both spatial and temporal explanations

Evaluation Application for Video Anomaly Detection 02/2021

- Project goal: Create a Python application that visualizes four criteria for video anomaly detection, namely frame-level, pixel level, track-based and region-based anomaly along with ROC curve
- Project outcome: Created an application that fulfills the requirements, supports 8 different datasets and extending options at optimized processing speed with multiprocessing and a friendly GUI
- Project methods: Built the application with PyQT5; fully tested with pytest for both unit testing and integration testing; optimized processing speed with multiprocessing Python package; implemented anomaly detection criteria closely based on StreetScene paper and created mock detection to proof test the evaluation algorithms given an AUC score.

Research Papers

Toward a Causal PM2. 5 Concentration Forecasting by Inferencing from Local Vehicle Tracking with a Low-Cost End-to-End Sensor System, 02/01/2026
Artificial Intelligence and Applications

When pandemics meet climate risk: An agent-based model of non-linear macroeconomic dynamics under compound stress, 12/01/2025
Journal of Economic Dynamics and Control

In-Context Learning for Data-Driven Censored Inventory Control, 12/01/2025
arXiv preprint arXiv:2605.14840

CliMaPan-Lab: An open-source Python framework for agent-based macroeconomic simulation of climate-and pandemic-related systemic risks, 12/01/2024
SoftwareX

Evaluating climate-related financial policies' impact on decarbonization with machine learning methods, *Scientific Reports* 12/01/2024

AlertTrap: A study on object detection in remote insects trap monitoring system using on-the-edge deep learning platform, 12/01/2024
Journal of Computational and Cognitive Engineering

Belief-Aware Inventory Control with Deep Mixture Models, 12/01/2024
NeurIPS 2025 Workshop MLxOR

PHEATPRUNER: Interpretable Data-centric Feature Selection for Multivariate Time Series Classification through Persistent Homology, 12/01/2024
arXiv preprint arXiv:2504.18329

Persistent Homology-induced Graph Ensembles for Time Series Regressions, 12/01/2024
arXiv preprint arXiv:2503.14240

Scalable, low-cost, and versatile system design for air pollution and traffic density monitoring and analysis, *Proceedings of the 8th International Conference on Sustainable Urban Development (ICSUD 2022)* 12/01/2023

Gesture Recognition Model with Multi Tracking Capture System for Human-Robot Interaction, *2023 International Conference on System Science and Engineering (ICSSE)* 06/01/2023

SmartTrap: An On-Field Insect Monitoring System Empowered by Edge Computing Capabilities, *2023 RIVF International Conference on Computing and Communication Technologies (RIVF)* 11/01/2023

Harnessing Sheaf Theory for Enhanced Air Quality Monitoring: Overcoming Conventional Limitations with Topology-Inspired Self-correcting Algorithm, 12/01/2022
Proceedings of the Future Technologies Conference (FTC) 2023

TSEM: Temporally-Weighted Spatiotemporal Explainable Neural Network for Multivariate Time Series, *Advances in Information and Communication: Proceedings of the 2023 Future of Information and Communication Conference (FICC)* 12/01/2022

A PM2.5 concentration prediction framework with vehicle tracking system: From cause to effect, *2022 RIVF International Conference on Computing and Communication Technologies (RIVF)* 11/01/2022

AlertTrap: On Designing An Edge-Computing Remote Insect Monitoring System, *2021 8th NAFOSTED Conference on Information and Computer Science (NICS)* 11/01/2021

Accuracy improvement of various short-term load forecasting models by a novel and unified statistical data-filtering method, *International Journal of Green Energy* 12/01/2019

A statistical data-filtering method proposed for short-term load forecasting models, *Journal of Electrical Engineering & Technology* 12/01/2019

Applying statistical analysis for assessing the reliability of input data to improve the quality of short-term load forecasting for a Ho Chi Minh City distribution network, *Science & Technology Development Journal-Engineering and Technology* 12/01/2018

Additional Information

- English, Vietnamese and German (Beginner)